

Durgesh Polavarapu

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Summary

Python Full Stack Developer with 3+ years of experience designing, building, and deploying scalable, secure, and performant web applications. Expert in modern backend frameworks Django, Flask, FastAPI, frontend technologies React, Angular, Vue, cloud infrastructure AWS, Azure, GCP, containerization (Docker, Kubernetes), and automated CI/CD pipelines. Proven ability to architect microservices and serverless applications, optimize database performance, and implement robust security measures. Passionate about writing clean, maintainable code and delivering business-critical software in Agile environments.

Experience

Python Full Stack Developer | Northern Trust | Remote

Jan 2025 – Present

- Architected microservices backend using FastAPI and Django REST Framework, boosting scalability and reducing downtime by 40%.
- Developed high-performance React.js frontend apps with TypeScript, reusable components, hooks, and Redux Toolkit for efficient state management.
- Designed GraphQL APIs with Apollo Server, enabling efficient data fetching and real-time updates through subscriptions.
- Deployed containerized apps on Kubernetes EKS with Helm charts; implemented zero-downtime CI/CD pipelines via GitHub Actions and Terraform IaC.
- Integrated OAuth2-based SSO and MFA with Keycloak, enforcing robust RBAC policies for enterprise-level SaaS apps.
- Optimized PostgreSQL schemas, queries, and indexing strategies, reducing data retrieval latency by 50%.
- Managed asynchronous processing with Celery and RabbitMQ, supporting millions of daily transactions with reliable recovery.
- Automated unit and E2E tests with Pytest and Cypress, maintaining >90% code coverage and reducing production bugs by 30%.
- Collaborated with cross-functional teams in Agile Scrum environments, facilitating sprint planning, code reviews, and documentation.

Python Full Stack Developer | Wipro | India

Jan 2021 – Dec 2022

- Built RESTful APIs and server-side apps using Flask, SQLAlchemy, and PostgreSQL for data analytics dashboards serving 50k+ users.
- Created interactive, responsive frontends using Vue.js and Bootstrap 4, improving UI accessibility and mobile performance.
- Migrated legacy monoliths to microservices with Docker and Kubernetes, significantly enhancing system scalability and maintainability.
- Integrated Kafka event streaming for real-time analytics pipelines, reducing data processing latency by 70%.
- Automated builds, tests, and deployments with Jenkins, Docker, and AWS ECS/Lambda.
- Maintained API docs with Swagger/OpenAPI for seamless frontend and mobile team integration.
- Implemented caching with Redis and CDN CloudFront, improving load times and reducing infrastructure costs by 25%.
- Used BDD frameworks Behave, Cucumber to align development with business requirements and increase test reliability.

Skills

Programming Languages: Python, JavaScript (ES6+), TypeScript, SQL, Bash, HTML5, CSS3, JSON, YAML

Backend Frameworks: Django, Django REST Framework, Flask, FastAPI, Tornado

Frontend: React.js, Angular, Vue 3, Next.js, Nuxt.js

State Management: Redux Toolkit, MobX, Recoil, Vuex

Styling: CSS3, SASS/SCSS, TailwindCSS, Bootstrap 5, Material-UI, Styled Components

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, TimescaleDB

ORMs & Migrations: Django ORM, SQLAlchemy, Alembic, Peewee

API Protocols: REST, GraphQL, gRPC, WebSockets

Auth & Security: OAuth2, OpenID Connect, JWT, RBAC, CSRF, CORS, OWASP Standards, HTTPS/SSL

Async & Messaging: Asyncio, Celery, RabbitMQ, Kafka, Redis Pub/Sub, AWS SQS/SNS

Cloud & DevOps: AWS, Docker, Kubernetes, Terraform, Serverless Framework

CI/CD: GitHub Actions, GitLab CI, Jenkins, ArgoCD

Testing & QA: Pytest, unittest, Selenium, Playwright, Jest, Cypress, React Testing Library, Cucumber

Monitoring & Logging: Prometheus, Grafana, ELK Stack, Sentry, Datadog, New Relic

Code Quality: SonarQube, ESLint, Prettier, Black, Flake8, Mypy

Version Control: Git, GitHub, GitLab, Bitbucket

Collaboration: Jira, Confluence, Slack

Others: Microservices, Serverless Architectures, Event-driven Architecture, Docker Compose, Agile Scrum, API Documentation

Education

Master's in Computer Science | Texas State University | Texas, USA

May 2025